Muhammad Mostafa Saadawy Amin

Junior Data Engineer

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PROFESSIONAL SUMMARY

Junior Data Engineer and final-year Computers & AI student with hands-on experience building scalable batch and streaming ETL pipelines in Python, Spark, Kafka, and Airflow.

Skilled in designing Star Schema data warehouses, improving query performance by 30–40%, and deploying analytics solutions on Microsoft Azure.

Strong problem-solving skills and analytical thinking developed through independent, end-to-end project work, paired with clear communication and interpersonal skills built through effective collaboration with cross-functional and academic teams. Focused on real-time data processing and CI/CD automation at scale.

PROFESSIONAL EXPERIENCE

Huawei Big Data Associate Trainee | Huawei ICT Academy *July 2026 – Sep 2026*

- Built foundational skills in Linux (command line, file systems, permissions), SQL query formulation, and Python (data structures, functions) using Jupyter Notebook for data processing tasks.
- Studied HDFS distributed storage architecture, ZooKeeper cluster coordination, and MapReduce/YARN for distributed batch processing and resource management.
- Worked Explored the Hadoop ecosystem including Hive (data warehousing), HBase (NoSQL columnar storage), and Elasticsearch (search/indexing) alongside Spark's in-memory computing model and Flink's unified stream-batch engine.
- Covered real-time data ingestion tools Flume (log aggregation) and Kafka (distributed messaging) within a complete Big Data pipeline architecture.

Microsoft Data Engineer – Trainee (DEPI Scholarship) | MCIT *Jun 2025 – Dec 2025*

- Built 10+ batch and streaming ETL pipelines using Python, SQL, Spark, Kafka, and Airflow for data ingestion and transformation.
- Designed 5+ Star Schema data warehouses and optimized analytical queries, improving performance by 30–40%.
- Processed millions of records using Hadoop, Hive, and Spark, implementing scalable Big Data processing workflows.
- Developed cloud data solutions using Azure Synapse, Azure Data Factory, and Azure Stream Analytics, with GitHub Actions for CI/CD automation.

Machine Learning Trainee | National Telecommunication Institute (NTI) *Aug 2025 – Sep 2025*

- Trained and evaluated supervised and unsupervised models using Python, NumPy, Pandas, and Scikit-learn on datasets.
- Applied data preprocessing, feature engineering, and model tuning to improve model performance and accuracy.
- Built end-to-end ML workflows from data collection and cleaning to model training, evaluation, and deployment via REST APIs using Streamlit.

PROJECTS

Big Data Project – Distributed Big Data & ML Platform

GitHub | Live Demo

- Built an end-to-end distributed Big Data platform for NYC FHVHV trip duration prediction using a 4-device K3s Kubernetes cluster connected through a private Tailscale mesh network.
- Implemented distributed ETL and ML workflows with Apache Spark/PySpark, leveraging 3 worker nodes with 32 CPU cores and ~20 GiB RAM for scalable data processing.
- Designed a Bronze/Silver/Gold Data Lake architecture using MinIO and orchestrated ingestion, transformation, feature engineering, and ML workflows with Apache Airflow.
- Integrated Apache Kafka, Spark Thrift Server, PostgreSQL, JupyterLab, and Streamlit to support streaming infrastructure, BI connectivity, interactive analysis, and analytics visualization.

Technologies: Kubernetes (K3s), Apache Spark, PySpark, Apache Airflow, Apache Kafka, MinIO, Python, PostgreSQL, Docker, Tailscale, Streamlit

IKEA Shopping – Admin Panel + Data Engineering & ML Pipeline *Apr 2026 – Jun 2026*

GitHub | Live Demo

- Extended a Java Swing admin panel (SQL Server, 10-table normalized schema, RBAC, SHA-256 hashing) with a full data engineering and ML layer, without modifying any existing Java code.
- Built a synthetic data generator producing 18 months of realistic sales data with monthly seasonality, weekly patterns, and growth trend using NumPy and pyodbc.

- Designed and implemented an ETL pipeline (Extract from SQL Server → Transform into a Star Schema → Load into SQLite warehouse) with Fact_Sales, Dim_Product, and Dim_Date tables.
- Trained an XGBoost forecasting model with lag features (lag-1, lag-7), 7-day rolling mean, and chronological train/test split; deployed a live Streamlit dashboard showing actual vs. forecasted daily sales per product, applying creative problem-solving to handle sparse and seasonal data.

Technologies: Java, Swing, SQL Server, Python, Pandas, NumPy, XGBoost, Streamlit, SQLite, ETL, Data Warehousing

Smart City – Real-Time Data Pipeline *Dec 2025 – Apr 2026*
GitHub
- Built an end-to-end real-time data pipeline simulating 5 smart city IoT sources (vehicles, GPS, traffic cameras, weather, emergencies) using Python producers and Apache Kafka.
- Processed concurrent Kafka streams with Apache Spark Structured Streaming, applying per-topic schemas, JSON deserialization, and watermarking to handle late-arriving events.
- Containerized the full distributed system (Zookeeper, Kafka broker, Spark master + 2 workers) using Docker Compose for one-command, reproducible deployment.
- Modeled data entities and relationships via ERD across 5 interconnected tables to support downstream analytics.

Technologies: Python, Apache Kafka, Apache Spark Structured Streaming, Docker, Docker Compose, Apache Parquet

Road Collisions Severity Prediction – ML Project *Sep 2025 – Nov 2025*
GitHub | Live Demo
- Built an end-to-end ML system to predict road collision severity (92.6% accuracy) using real-world traffic accident data.
- Performed data cleaning, feature engineering, and EDA to identify key factors affecting accident severity.
- Trained and evaluated a CatBoost classifier, achieving a 0.92 weighted F1-score with perfect precision on Fatal and Serious collision classes.
- Deployed an interactive Streamlit web application for real-time severity predictions.

Technologies: Python, CatBoost, Scikit-learn, Pandas, Streamlit, Matplotlib, Seaborn

TECHNICAL SKILLS

Programming Languages: Python, SQL, C++, Java, C#
Data Engineering: ETL/ELT Pipelines, Data Modeling, Data Warehousing, Star Schema, Batch Processing, Streaming Processing, Data Validation
Big Data & Distributed Processing: Apache Spark, PySpark, Apache Kafka, Hadoop, Hive, Spark Structured Streaming, Apache Flink, HBase, Elasticsearch
Data Orchestration & Automation: Apache Airflow, dbt, GitHub Actions, CI/CD, Apache ZooKeeper, Apache Flume
Cloud & Azure: Azure Synapse Analytics, Azure Data Factory, Azure Databricks, Dedicated SQL Pools, Azure Stream Analytics
Databases & Data Storage: PostgreSQL, MySQL, SQL Server, SQLite, MongoDB, ClickHouse, Delta Lake, MinIO
Distributed Systems & DevOps: Kubernetes (K3s), Docker, Docker Compose, Tailscale, Git, GitHub, Linux
Data Analytics & ML: Pandas, NumPy, Scikit-learn, CatBoost, NLTK, Matplotlib, Seaborn
Development & BI Tools: Jupyter Notebook, JupyterLab, Streamlit, REST APIs, Spark Thrift Server, Pipeline Monitoring
Human Languages: Arabic (Native), English (Professional Working Proficiency)

EDUCATION

Bachelor of Computers and Artificial Intelligence *Expected: 2027*
Minia National University, Egypt
Relevant Coursework: Data Structures, Data Mining, Algorithms, Database Systems, OOP, Machine Learning

LEADERSHIP & ACTIVITIES

- Mentor at ICPC MNU Team — guided competitive programming students in problem-solving and algorithms, strengthening peer interpersonal and coaching skills.
- Mentor at MENTOR Network — supported junior students in career development and technical growth.
- Admin at Student Activity — managed organizational and academic community events, collaborating with student committees.